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Low Complexity QCA-Based 1:2^n Demux Architecture With Effective Modular Design

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ABSTRACT

Quantum-dot cellular automata (QCA) is a revolutionary technology operating at the nanoscale and having the ability to supplant the CMOS technology. Demultiplexers (DeMuxs) are essential combinational blocks in receiver-side communication systems, where they perform signal distribution and serial-to-parallel data routing. This paper introduces a low integration complexity, coplanar 1:2 DeMux circuit utilizing 27 QCA cells, designed to provide direct access to all inputs and outputs and enable straightforward modular expansion. Based on this design, efficient 1:4 and 1:8 DeMux circuits have been proposed. The performance of the designed DeMux circuits is evaluated and compared with existing designs with the help of QCA Designer 2.0.3 in terms of area, cell count, latency, and energy dissipation. Results show architectural- and layout-level advantages over existing layouts. Energy analysis using QCADesigner-E shows average energy dissipation values of 7.92, 17.2, and 55.9 meV per cycle for the 1:2, 1:4, and 1:8 DeMuxs, respectively, demonstrating the effectiveness of the proposed modular architecture.

1 | Introduction

Traditional transistor-based semiconductor devices are increasingly facing challenges with scaling. As supply voltages decrease, the leakage current makes the power consumption a significant issue in transistor circuits. According to Moore's law, the feature sizes of the CMOS are approaching physical limits in the coming decades. The reduction in the size of CMOS leads to serious issues such as increased leakage current, higher complexity of design, and substantial lithography costs. In response, alternative nanotechnologies have garnered considerable attention in recent years. Lent et al. [1] in 1993 introduced the concept of quantum-dot cellular automata (QCA) for circuit design . This approach has since evolved, with electronic QCA and magnetic QCA emerging as highly promising nano IC devices [2, 3]. In contrast to the CMOS technology, QCA is working by means of a Coulombic repulsion between electrons allowing computations without current flowing resulting in an order of magnitude more power efficient. This basic difference makes QCA an attractive option for the future integrated circuits, and able to surpass the limitations of conventional CMOS technology.

Different implementations of bistable mechanisms as well as local interactions compatible with the QCA paradigm have been proposed till date. These realizations range from atomic and molecular QCA designs based on the electrostatic interaction to structures made of semiconductors. A few devices in QCA [4, 5] have been fabricated and experimentally tested, which demonstrate the feasibility as well as potential of these approaches. QCA technology has developed, and many

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different circuits [6–8] have been researched in recent years. One major challenge of circuit design is dealing with circuit parameters like area, complexity, and clock delays. Minimizing these parameters are central concerns in theories being developed for new QCA modules. In modern processor architectures, efficient data routing and control signal distribution are critical for the implementation of subsystems such as instruction decoding units, register files, interconnection networks, and pipeline control logic. QCA-based logic blocks, including multiplexers and DeMuxs, play a fundamental role in realizing these functions at the nanoscale by enabling low-power, high-density signal steering without current flow. In receiver-side communication systems, DeMuxs are typically employed to distribute received data or control signals to parallel processing blocks such as decoding, buffering, or control units. In this context, the proposed QCA-based DeMux is intended to function as a nanoscale signal-routing and distribution block, rather than as a complete receiver architecture. It serves as a serial-to-parallel converter and forms a significant component in arithmetic logic units (ALUs). It finds extensive application in signal routing, data communications, and data bus control. Various configurations of QCA cells enable a wide range of DeMux designs focusing on improving their performance.

This study concentrates on the layout design and simulation evaluation of QCA DeMuxs. Although practical fabrication and implementation are important, they are considered as future work, as successful layout and clocking optimization are prerequisites for reliable experimental realization. Although significant research has been conducted on QCA-based designs, the focus on DeMux circuit design remains relatively limited in the literature. Some existing designs achieve lower local metrics by aggressively minimizing cell count or area; however, such designs often require extra cells, crossovers, or layout modifications when connected to other QCA circuits or extended to higher-order structures. Therefore, optimized and modular QCA DeMux designs, such as the one proposed in this work, contribute directly to the scalable construction of processor-oriented QCA circuits. Here, the primary focus is on introducing an efficient 1:2 DeMux using QCA technology. The proposed design emphasizes integration, scalability and physical realizability rather than mere area and delay minimization. This 1:2 DeMux can be further used as the basic building block in making up 1:4 and 1:8 DeMux.

Detailed comparisons with other relevant designs highlight significant improvements in terms of cell count and area efficiency in the presented designs. As a result, the objectives of this investigation are as follows:

- Introducing an optimal 1:2 DeMux design that enable quick interfacing and offer a modular structure.
- Design and analysis of higher order DeMuxs such as 1:4 and 1:8 DeMux based on the proposed 1:2 DeMux structure.
- Comparing with similar existing designs, we evaluate common parameters such as cell count, complexity, delay, and area.
- Presenting validity of the suggested layouts is confirmed through rigorous simulation using the QCA Designer tool.

This paper is organized as follows: Section 2 introduces the QCA basics; Section 3 presents existing DeMuxs and research gaps; proposed DeMuxs are showcased in Section 4; Section 5 provides an analysis of the proposed designs and compares it with others; finally, Section 6 presents the conclusion of this study.

2 | QCA Background

The fundamental building block of QCA technology is the quantum cell, composed of four minuscule quantum dots. The cell is of the square shape and the quantum dots are positioned at the edges of this square shaped cell. The quantum cell accommodates two electrons and these two electrons can freely move to quantum dots. Thus, due to the influence of Coulombic repulsion, the two electrons remain at the diagonal positions in the QCA cell [9]. Although these electrons are able to swap positions within a cell, such exchanges are not allowed between the adjacent QCA cells. On the basis of the orientation of these electrons, a QCA cell can exhibit one of two polarizations, denoted as $P = +1$ representing logic "1" state and $P = -1$ expressing logic "0" state.

Figure 1 shows the QCA cell and the mentioned two configurations of the cell.

In QCA technology, a QCA wire comprises adjacent cells aligned in series.

The QCA wire can be designed using simple (90°) QCA cells or rotated (45°) QCA cells, as illustrated in Figure 2.

The fundamental components of every QCA circuit are the inverter and the majority gate.

Figure 3 depicts a schematic of the various inverters, which reverses the input signal's value.

The majority gate produces the output in accordance with the status of the majority of the inputs.

Additionally, Figure 4 shows the schematic of the 3-input Majority Gate—that is, original majority voter/gate (OMG) and rotated majority gate (RMG). This gate performs the logic function $M(a, b, c) = ab + bc + ca$. The polarity of the device cell depends on the majority of input cells polarity. Further, the output cell attains the polarity in accordance with the polarity of the device cell. Notably, this function can be easily transformed into a 2-input OR or AND gates by connecting one of the three inputs to logic "1" or logic "0", respectively. Most of the logic circuits in this technology are implemented with the use of the inverters and 3-input majority gates.

However, the inverter gate and the 3-input majority gate have been fundamental in previous circuit designs, but researchers are actively exploring new logic gates aimed at improving circuit efficiency.

In QCA circuits, signal synchronization is achieved using a four-phase clocking scheme [10].

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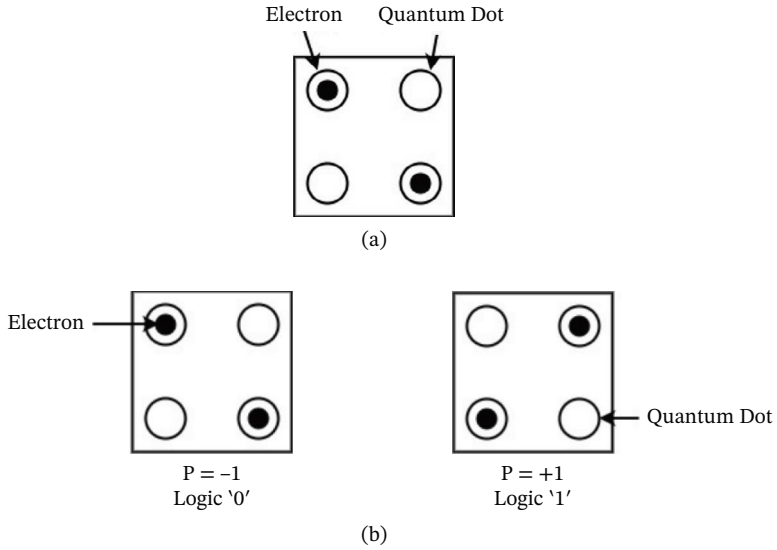


FIGURE 1 | (a) Quantum cell and (b) polarization of cell.

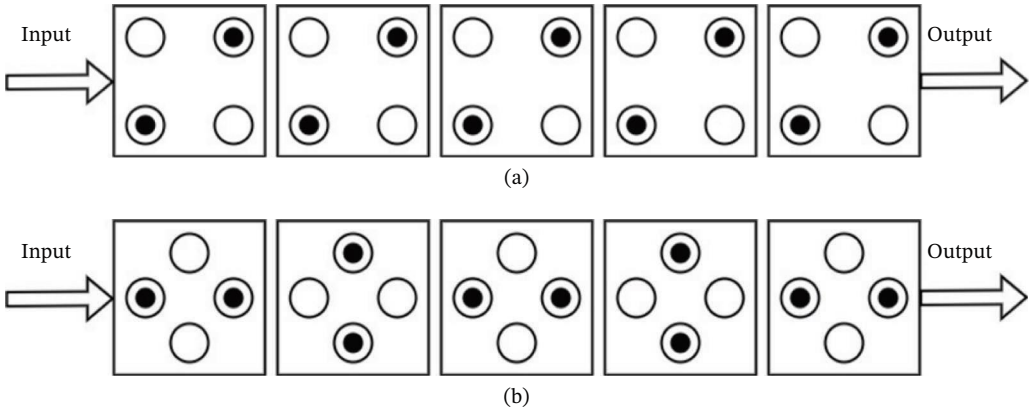


FIGURE 2 | QCA wires (a) 90° binary wire; (b) 45° binary wire.

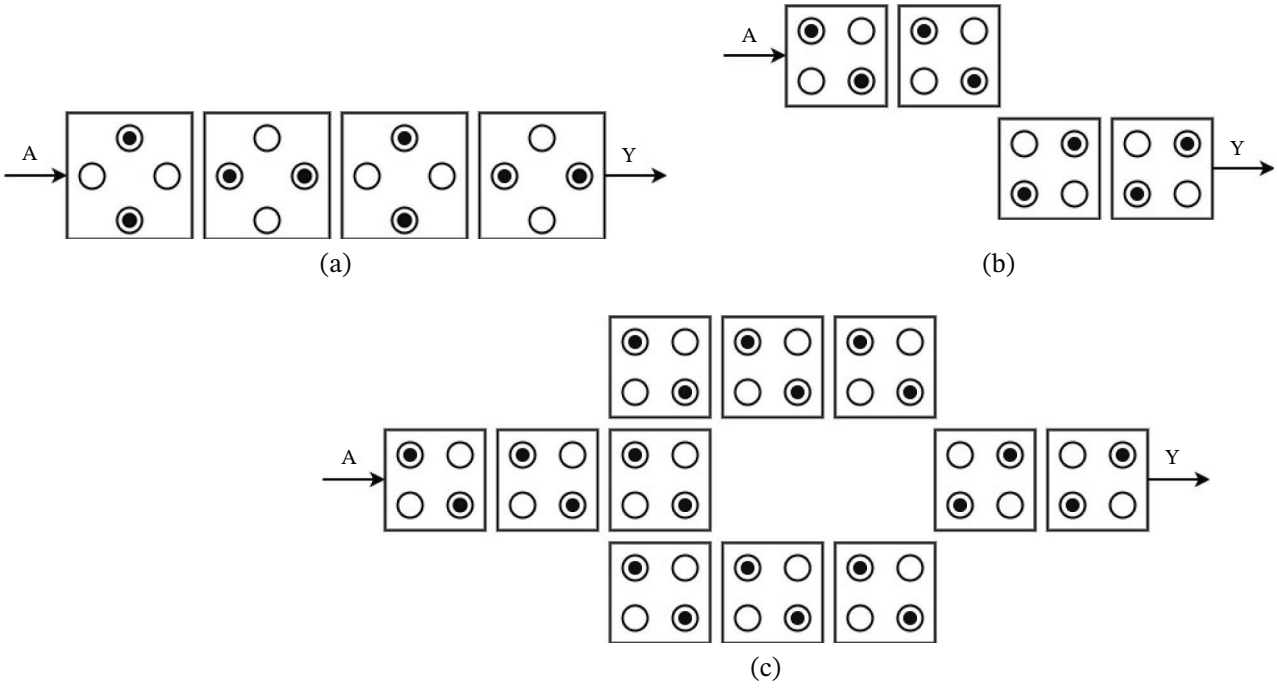


FIGURE 3 | Inverter gate with (a) 45° cells, (b) simple gate, and (c) robust gate.

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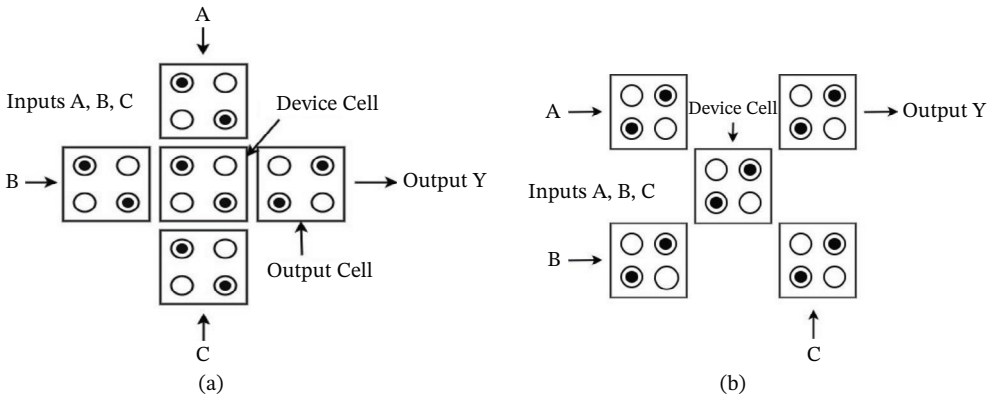


FIGURE 4 | 3-Input majority gate: (a) original majority gate and (b) rotated majority gate.

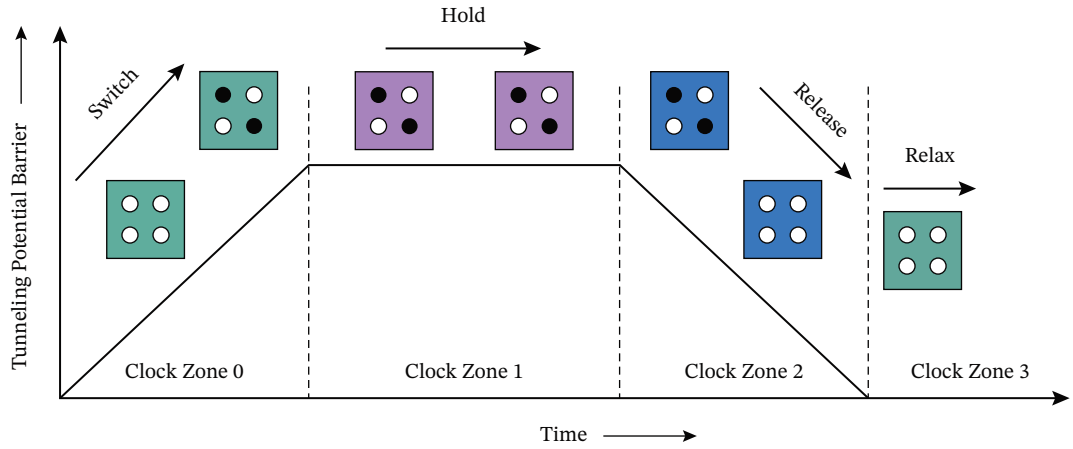


FIGURE 5 | QCA Clocking.

This scheme includes four different phase switches, hold, release, and relax, each spaced 90 separated, as depicted in Figure 5. One complete clock cycle corresponds to four clock phases. In this work, delay is primarily reported in terms of clock cycles.

This clocking mechanism ensures reliable switching and prevents signal feedback towards the input direction. Furthermore, leveraging the phase difference properties of QCA clocking allows for the realization of coplanar crossings.

In QCA, wire crossings can be categorized into two types: single-layer wire crossing and multilayer wire crossing. The single-layer wire crossing also known as coplanar wire crossing can be designed with the incorporation of the phase difference of 180 between the wires, as depicted in Figure 6a. Such type of wire crossing is also called as logical wire crossing. Alternatively, an inverter chain can be used for wire crossing without requiring any phase difference, as shown in Figure 6b.

The proposed QCA-based DeMux employs coplanar crossings to enable compact and efficient interconnections within the circuit. Coplanar crossings have been shown to be feasible in QCA designs, as demonstrated in prior simulation studies [11, 12], where careful clock zone assignment and polarization control allow signals to cross without interference. Though manufacturing yield and alignment precision are crucial factors for realizing QCA devices physically, these lie outside the scope of this work, which provides a logical design and simulation of the DeMux. Other work has focused on the fabrication concerns surrounding cell misalignment and defects [5, 13], but our work focuses on elucidating circuit functionality, layout viability, and performance at a design level under perfect cell placement and operation.

Conversely, multilayer wire crossing will demand at least three of the design layers as illustrated in Figure 6c. Although the multilayer crossings allow for compact circuit designs, they are difficult to realize physically. Hence, crossing in a single layer is usually much more realizable for physical embedding. Single-layer wire crossings have been used in all the circuits presented here.

Clock congestion and skew in conventional CMOS systems offer a performance ceiling due to the need for complex global clock trees, which increase latency and power consumption as well as sensitivity to process, voltage and temperature variations; this imposes an upper bound on effective operating frequency. QCA overcome these challenges using localized multiphase clocking zones that supply timing on demand while actively regulating computation and signal propagation by avoiding long-range routing of the clock, which can potentially achieve a drastic reduction in skew. This method supports pipelined barrier computation and fine-grained synchronization making the circuits more compact and energy-efficient. However, the performance

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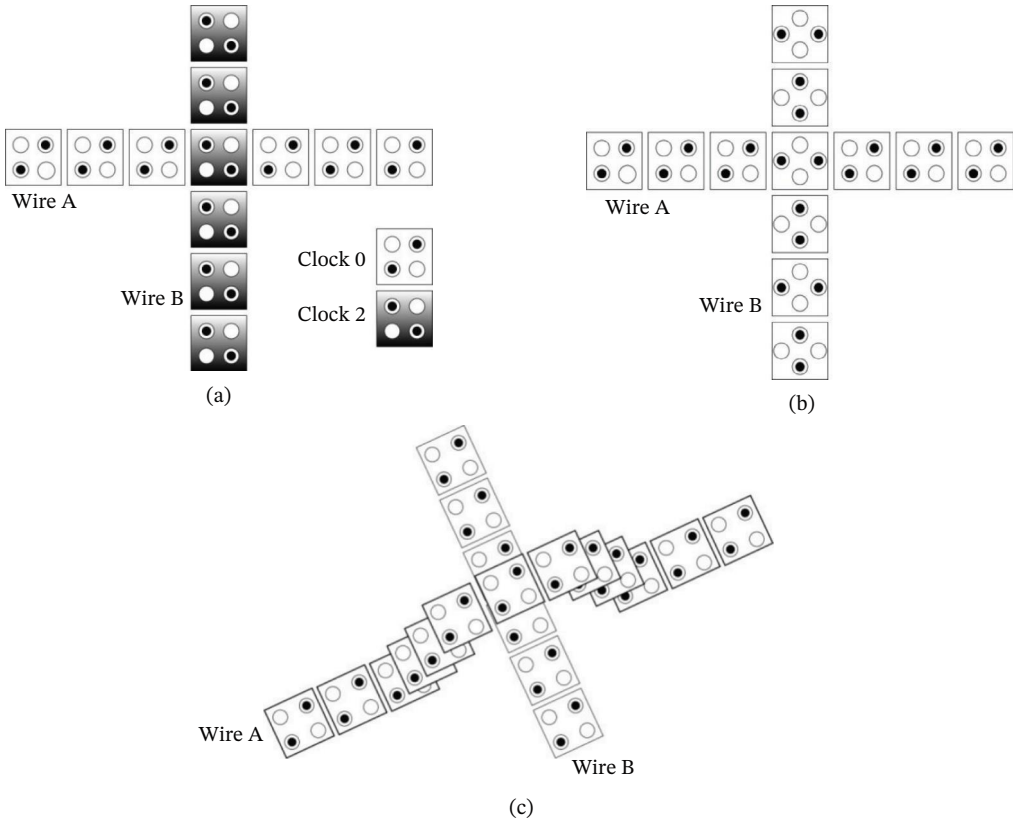


FIGURE 6 | QCA wire crossing: (a) coplanar logical crossing, (b) inverted chain crossing, and (c) multilayer crossing.

of QCA still remains limited by physical and architectural factors such as interdot tunneling rates, adiabatic switching conditions, clock slope, and clock zone layout which together set the maximum operating frequency that can be achieved. Clock partitioning gone awry or excessive clock zone size can lead to latency, fan-out restriction, or signal synchronization issues at the domain level. So even though QCA alleviated classical clock congestion and skew, it introduces new upper limits on the basis of device physics, as well as clocking dynamics and layout considerations, reiterating that careful clock and architectural design is critical to scalable, high performance QCA circuits.

3 | Related Work on QCA DeMux and Research Gaps

The QCA DeMux holds significant importance in the fields of nanocommunication and nanocomputation. However, unlike multiplexers, there are limited instances of QCA DeMuxs documented in below mentioned recent literature. Various architectures and implementations of QCA-dependent DeMuxs have been explored in this section. A key strength of this work lies in its comparison of the proposed DeMux with several state-of-the-art existing designs.

In 2017, Das and De [14] introduced a 1:2 DeMux as part of a circuit switching network designed using QCA technology. Their DeMux design utilized 32 cells, operated with three clocks, incorporated one inverter, and employed three majority gates. The total area occupied by their DeMux design was $0.026\mu m^2$. Although the design achieves correct functionality, the allocation of only one cell per clock zone in certain regions may reduce polarization stability and increase susceptibility to fabrication defects and thermal noise, thereby affecting reliability. Furthermore, the architecture functions with relatively elevated clock latency without capturing energy dissipation analysis under a uniform simulation model.

Sharma and Kaushik [15] proposed a small 1:2 QCA DeMux structure. The 1:2 DeMux based design using 11 QCA cells with an area of $6480nm^2$. It was reported that a 16 to 256 DeMux in this design has a latency of only one clock phase. The layout has better area and delay figures than standalone 1:2 DeMux. This cascade to high-order DeMux (1:4, 1:8, etc.) is related with a slight appreciation of routing complexity, wire crossings, and management of clock zone that could make some scalability, layout regularity, and robustness quite poor.

Afrooz and Navimipour [16] developed a 1:2 multilayer DeMux using QCA technology. Their design employed 21 cells and occupied a covered area of $0.017\mu m^2$. The DeMux operated with two clock phases, incorporating one inverter and two majority gates into its structure. Further, they extend this to design 1:4 QCA-based DeMux. The designs present several challenges, such as direct interfacing not possible in cascading system, because all input and output cells are not accessible directly. To overcome, these designs need additional cells and crossovers. Furthermore, designing multilayer layouts with QCA technology poses significant challenges due to its inherent complexity compared with single-layer designs.

Ahmad [17] proposed a QCA-based MUX/DeMux architecture. Specifically, their design for a 1:2 DeMux occupied a total area

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of 0.023 μm^2. The design utilized 21 QCA cells and achieved a delay of 0.5 clock cycles, as reported in the paper. With the use of two clock phases, the DeMux was implemented with two RMGs and one inverter. Furthermore, the author demonstrated the applicability of their approach by extending it to design an efficient 1:4 DeMux. This configuration utilized two inverters and four 3-input AND gates, demonstrating the scalability of their QCA-based DeMux design. The mentioned designs present several challenges, including the design of nonuniversal 3-input AND gates and the issue of output cells not being aligned in the same direction. These factors contribute to increased complexity, especially when cascading multiple stages of these DeMux architectures.

Khan and Arya [18] introduced a straight forward and optimized single-layered DeMux based on QCA technology. This design utilized 21 cells with total area of 0.020 μm^2 and operated efficiently with two clock phases. The DeMux was engineered with a single Inverter and two Majority Voters. The two majority voters were used as AND gates. This coplanar configuration minimized the number of cells required. The design present challenges such as direct interfacing not possible in cascading system, because all input and output cells are not accessible directly. In addition, the design has only one cell per clock zone at the input, which can present reliability issues. To overcome these issues, designs need additional cells and crossovers, which in return increase area, cell counts, and delay.

Gassoumi et al. [19] developed a 1:2 coplanar DeMux using QCA technology. Their design employed 19 cells and occupied an area of 0.02 μm^2. The proposed DeMux operated with two clock phases and incorporated one inverter along with two majority gates. However, the design mainly focuses on standalone 1:2 DeMux implementation and does not demonstrate higher-order modular expansion. In addition, direct interfacing in cascading systems is difficult because all input cells are not directly accessible. Consequently, additional routing cells and crossovers are required for modular integration and higher-order realization.

In 2023, Asthana et al., [20] presented a QCA-based digital switch matrix architecture that includes a 1:2 DeMux. Their design occupied a total area of 0.023 μm^2 and utilized 20 cells, achieving a latency of 0.5 clock cycles. Operating with two clock phases, the DeMux was realized using two RMGs and one inverter. However, the design faces challenges such as output cells not aligning in the same direction. These factors contribute to increased complexity, particularly when cascading multiple stages of these DeMux architectures.

Sharma [21] in 2021, developed an optimal, coplanar, single-clock-phase 1:2 DeMux circuit that utilizes only 19 QCA cells. The compact structure occupies a QCA area of 0.014 μm^2. Higher order circuits—that is, 1:4 and 1:8 DeMux—were also designed with the help of the said 1:2 DeMux circuit. The designs face several challenges: (i) In the 1:2 DeMux structure, input *D* and *S* need to be fed in opposite directions, requiring additional multilayer crossovers and cells to align the inputs in the same direction. Coplanar crossovers may generate interference with neighboring cells. (ii) Direct access to input cells S0, S1, and S2 is not possible in the 1:4 and 1:8 DeMux configurations. iii) Designing multilayer layouts with QCA technology presents significant challenges due to its inherent complexity.

Abdullah-Al-Shafi [22] introduced a compact 1:2 DeMuX implemented without majority gates, with functionality achieved through direct interaction among cells. The given 1:2 DeMux design employing nine QCA cells, with total area of 0.01 μm^2. The latency of this DeMux design was reported to be one clock phase. The layout achieves superior performance in terms of minimum area and latency for a standalone 1:2 DeMux. However, extending it to higher-order DeMuxs introduces additional routing complexity, crossovers, and clocking challenges. Although the approach is area-efficient, the close placement of input, Out1, and Out2 cells increases intercell interaction, which may limit scalability and integration into larger circuits due to potential interference issues.

Vaid et al. [23] proposed a 1:2 DeMux design utilizing only 17 QCA cells with a single clock phase. The design also includes an analysis of power dissipation. It occupies a total area of 0.016 μm^2, with an effective cell area of 0.006 μm^2. The architecture is simple and robust as it employs a single-layer layout and avoids the need for crossover structures for a standalone 1:2 DeMux. However, the design presents certain limitations. First, the inputs *D* and *S* must be applied from opposite directions, which introduces complexity in signal routing and may require additional crossovers and cells. Second, when extending the design to higher-order DeMux configurations, it necessitates extra routing elements, crossovers, and layout modifications, thereby increasing design complexity and reducing scalability.

Akbari-Hasanjani and Sabbaghi-Nadooshan [24] proposed an optimized 1:2 DeMux implemented as a tree network switch using a RMG in QCA technology. The design employs 16 QCA cells, achieving a compact layout with a total area of 0.02 μm^2 and a latency of 0.25 clock cycles. However, the design presents certain limitations. First, the inputs O_0 and C must be applied from opposite directions, which complicates signal routing and may require additional crossovers and cells. Second, when extending the design to higher-order DeMux configurations, additional routing elements, crossovers, and layout modifications are required. This increases overall design complexity and limits its scalability.

Jeon and Changho [25] introduced a QCA-based 1:2 DeMux for communication and signal routing applications. The design is based on single-layer cell interaction, achieving efficient area utilization with a minimal number of cells and reduced footprint. The proposed unit DeMux is inherently scalable through the use of multilayer crossovers. Although the design demonstrates strong quantitative performance for a standalone 1:2 DeMux, cascading multiple 1:2 DeMux units leads to increased cell count and area overhead. Additionally, the close placement of input cell *A* and output cells may introduce interference issues. However, the design exhibits certain drawbacks. In particular, implementing multilayer layouts in QCA technology is challenging due to increased fabrication complexity compared with single-layer designs, which limits practical scalability.

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The previously discussed designs face the same problems, as described above. The important literature gaps can be summarized as follows:

- Create DeMuxs with fast interfacing and a modular structure.
- Mitigate complexity through careful layout and architectural design.
- Optimize performance metric, such as smaller area footprint and lower power consumption.
- Improve direct access to inputs and outputs leading to better signal integrity through several stages.

However, this study fills each of these research gaps by proposing the modular DeMux layout that is likely to enable a more straightforward design for higher-order DeMuxs. It is important to note that the proposed layout does not primarily target absolute minimization of cell count or area at the individual DeMux level. In contrast, the proposed design offers direct access to all input and output cells, adopts a single-layer coplanar layout (which eliminates the need for rotated cells or vertical shifts), allows simple "cascading" without any change of layout, and has predictable linear delay scaling with higher order $1{:}2^n$ DeMuxs. These attributes minimize integration overhead and eliminate extra routing cells and crossovers typically needed in the competition. When such modifications are accounted for, the effective area, cell count, and latency of many existing architectures increase beyond their reported standalone values.

4 | Proposed Designs

A DeMux, functioning as a data distributor, is a combinational circuit that routes a single input data signal to one of several output lines on the basis of the values of the select lines. In a $1{:}2^n$ DeMux, the circuit comprises m AND gates (where $m = 2^n$), one for each of the 2^n possible combinations of the n select lines. The single data input is fed to all the AND gates, but only one AND gate will be active at a time, determining which output line receives the input.

4.1 | 1:2 DeMux

A 1:2 DeMux, specifically, has two outputs and uses one select line to direct the input data, as depicted in Figure 7.

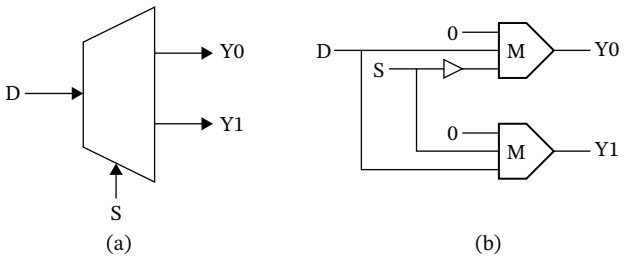


FIGURE 7 | 1:2 DeMux (a) block diagram and (b) schematic diagram.

As shown in the schematic diagram, constructing the 1:2 DeMux requires an inverter and two-input AND gates.

This design aims to enhance efficiency and reduce complexity. The following equations express the output of the 1:2 DeMux.

Output $Y0 = DS'$

Output $Y1 = DS$

The truth table of 1:2 DeMux is explained in Table 1.

Figure 8 depicts the QCA layout of the coplanar 1:2 DeMux presented here, designed to achieve high device density, minimal layout area, and improved latency. In the diagram, D represents the DeMux input, S denotes the select line, and $Y0$ and $Y1$ are the outputs of the DeMux. This DeMux design utilizes 90° QCA cells with a single crossover. The proposed layout utilizes only 27 QCA cells and occupies a compact area of $0.033\,\mu m^2$ (refer to Figure 8).

4.2 | 1:4 DeMux

The separate block diagram and schematic diagram of 1:4 DeMux is shown in Figure 9. The proposed QCA 1:2 DeMux can be used as primitive building blocks to generate the higher-order DeMux circuits.

As shown in the schematic diagram, 1:4 DeMux can be formed by integrating three modules of 1:2 DeMux.

TABLE 1 | Truth table of conventional 1:2 DeMux.

Inputs		Outputs	
S	***D***	***Y0***	***Y1***
0	0	0	0
0	1	1	0
1	0	0	0
1	1	0	1

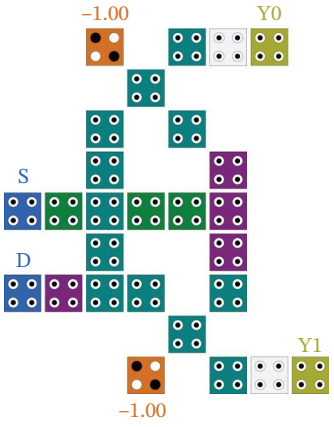


FIGURE 8 | Layout of proposed 1:2 DeMux.

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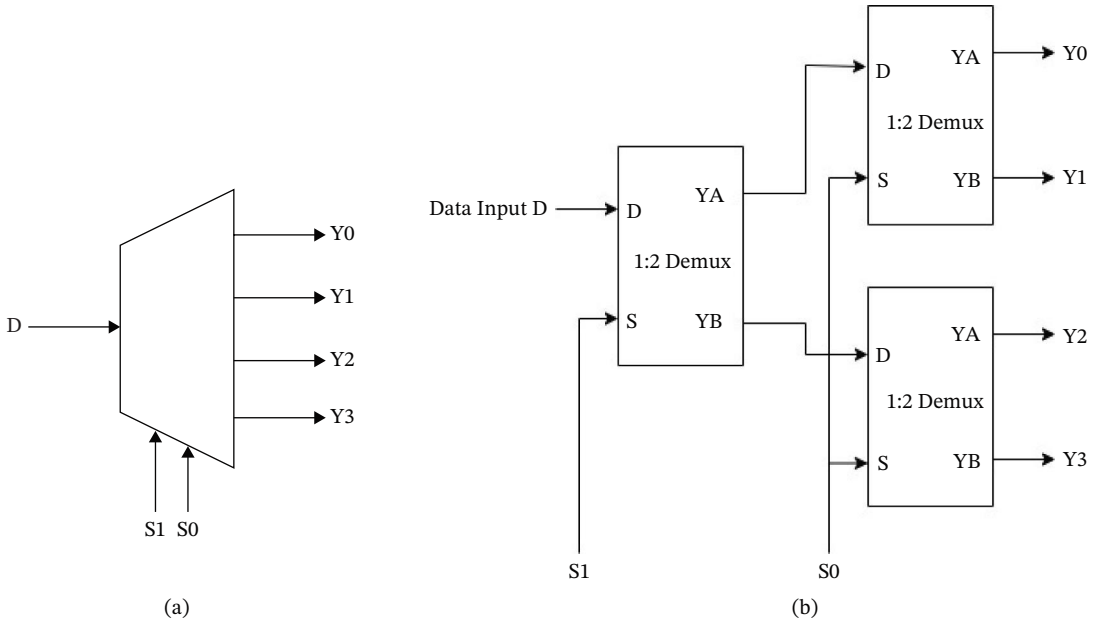


FIGURE 9 | 1:4 DeMux (a) block diagram and (b) schematic diagram.

TABLE 2 | Truth table of conventional 1:4 DeMux.

Inputs			Outputs			
S1	***S0***	***D***	***Y0***	***Y1***	***Y2***	***Y3***
0	0	0	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	1	0	1	0	0
1	0	0	0	0	0	0
1	0	1	0	0	1	0
1	1	0	0	0	0	0
1	1	1	0	0	0	1

This logic arrangement has each output from the first DeMux as an input for the next one. The design of the 1:4 DeMux involves two NOT gates and four AND gates, alongside the selection lines $S0$ and $S1$ which determine the routing of inputs to specific output lines.

Table 2 presents the truth table for the 1:4 DeMux, facilitating the derivation of its logic expression as shown in the following equation.

Output $Y0 = DS1'S0'$

Output $Y1 = DS1'S0$

Output $Y2 = DS1S0'$

Output $Y3 = DS1S0$

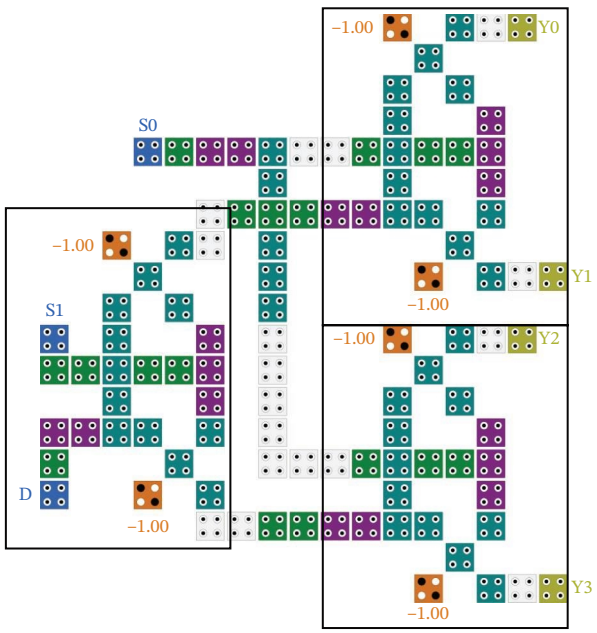


FIGURE 10 | Layout of proposed 1:4 DeMux using 1:2 DeMuxs.

Figure 10 shows the QCA design of the proposed 1:4 DeMux comprising blocks same as in the case of 1:2 DeMux. Various combinations of the selection lines ($S1$, $S0$) dictate to which output line the input will be routed. However, this 1:4 DeMux structure uses coplanar crossover wiring and conventional whole QCA cells to be deployed in a single plane.

The entire QCA structure comprises 105 QCA cells, occupying a layout area of $0.16\,\mu m^2$ (Figure 10). It requires eight clock phases (two clock cycles) to ensure the generation of valid outputs, as illustrated in Figure 11. This design ensures efficient operation and compact spatial utilization within nanoscale computing environments.

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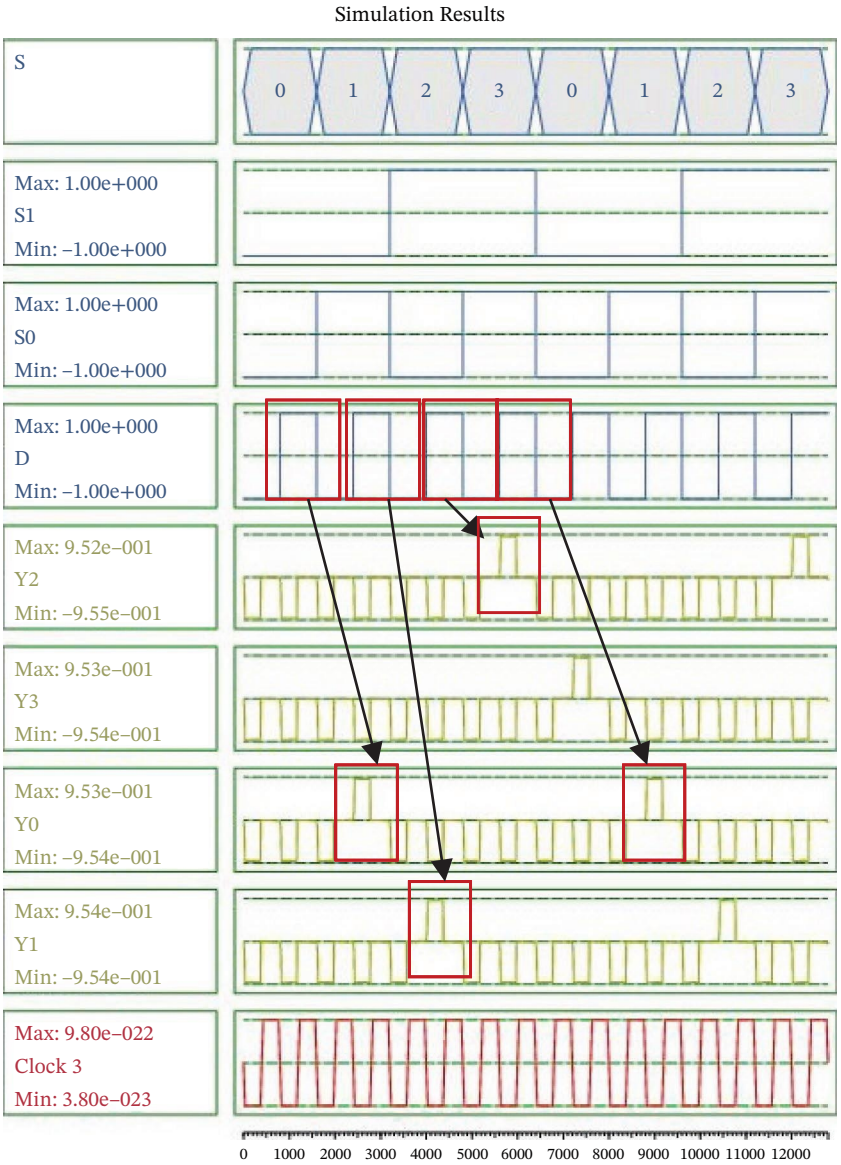


FIGURE 11 | Simulation waveform of the proposed 1:4 DeMux.

4.3 | 1:8 DeMux

The 1:8 DeMux circuit is built with the help of the proposed 1:2 DeMux design.

Figure 12 also shows the block diagram of the 1:8 DeMux. It requires seven proposed 1:2 DeMux circuits in such configuration. As shown in Figure 12, the system has one input line (*D*), three select lines (*S*0, *S*1, and *S*2), and eight output lines (*Y*0, *Y*1, *Y*2, *Y*3, *Y*4, *Y*5, *Y*6, and *Y*7). Depending on how selection lines *S*0, *S*1, and *S*2 have been set, the input signal will go to one of the output terminals (Refer to Table 3).

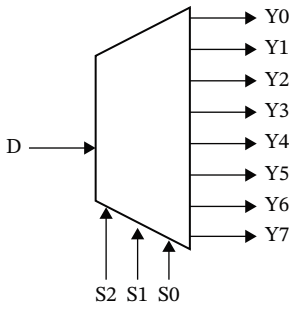


FIGURE 12 | 1:8 DeMux block diagram.

Table 3 outlines the logical operations of the 1:8 DeMux circuit, specifying how each combination of selection lines (*S*0, *S*1, and *S*2) directs input data to the appropriate output terminal.

Figure 13 shows the layout of the proposed 1:8 DeMux using QCA cells. The circuit has been designed on only 325 QCA cells covering an area of about 0.55 μm^2. This design works with a throughput of three clock cycles. In this case, a coplanar crossover approach is used between stages, where outputs of one stage are fed to the input of the next stage. This method uses cells in clock Zones 0 and 2 for crossover. The simple extensibility of the previous 1:4 DeMux makes it very easy to achieve the scale at a 1:8 DeMux level with minimal complexity. The proposed DeMux circuits have been analyzed with respect to latency quantitatively.

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TABLE 3 | Truth table of conventional 1:8 DeMux.

Inputs			Outputs							
S2	***S1***	***S0***	***Y0***	***Y1***	***Y2***	***Y3***	***Y4***	***Y5***	***Y6***	***Y7***
0	0	0	D	0	0	0	0	0	0	0
0	0	1	0	D	0	0	0	0	0	0
0	1	0	0	0	D	0	0	0	0	0
0	1	1	0	0	0	D	0	0	0	0
1	0	0	0	0	0	0	D	0	0	0
1	0	1	0	0	0	0	0	D	0	0
1	1	0	0	0	0	0	0	0	D	0
1	1	1	0	0	0	0	0	0	0	D

The measured delays for the 1:2, 1:4, and 1:8 DeMuxs are equal to one, two, and three clock cycles, respectively, in accordance with $n = 1,2,3$ as illustrated in Figures 8, 10, and 13. This means that we can be assured that the delay increases linearly with n, which matches our expectations based on the behavior of cascaded DeMux stages. Each added stage incurs one more clock cycle of delay.

4.4 | Modular 1:2^n DeMultiplexer Architecture

From the previous section, it is observed that the proposed design exhibits a linear relationship between latency and the number of stages, resulting in predictable timing behavior for higher-order circuits. Specifically, the delay of a 1:2^n DeMux is equal to n clock cycles. The proposed 1:2 DeMux acts as a fundamental modular building block for constructing higher-order 1:2^n DeMux architectures. These configurations are realized by hierarchically cascading identical 1:2 DeMux modules, where each additional select line introduces a new stage in the design. Consequently, a 1:2^n DeMux can be implemented using $(2^n - 1)$ identical 1:2 DeMux modules, ensuring predictable scaling in both latency and hardware complexity. This modular approach is validated through the presented 1:4 and 1:8 DeMux implementations. For instance, a 1:16 DeMux requires 15 $(2^4 - 1)$ modules, whereas a 1:32 DeMux requires 31 $(2^5 - 1)$ modules. In general, a 1:2^n DeMux with n selection lines can be constructed using $(2^n - 1)$ 1:2 DeMux units.

5 | Results & Discussion

This section provides detailed evaluation of the proposed DeMux circuit and the same has been compared against existing designs. Since the QCADesigner 2.0.3 and its QCADesigner-E tool was used for the analysis, the default bistable approximation settings were taken as a baseline [10]. The characteristics assessed include crossover, clock cycle, layout cost, cell count, and total area covered.

Figure 14 illustrates the simulation results for the proposed QCA-based 1:2 DeMux, confirming its functionality. The results show that when the selection Line S is low, Input D is routed to Output $Y0$, whereas when S is high, Input D is routed to Output $Y1$. Thus, varying the selection Line S properly directs Input D to either $Y0$ or $Y1$. Both Outputs $Y0$ and $Y1$ are correctly generated after a delay of one clock cycle, as shown in red boxes in Figure 14. It is worth noting that, the correct outputs are generated at the hold phase of Clock 3. For simplicity, only Clock 3 is shown in the simulation result of the proposed 1:2 DeMux.

The implementation involved two RMGs and one inverter, highlighting its efficient use of QCA components for demultiplexing operations. Furthermore, it requires four clock phases to ensure the generation of valid outputs, as illustrated in Figure 14. With this modular working, higher order DeMux circuits can be built very conveniently. The design specifications are as follows:

- Cells no (count): 27
- Size of cell: 18×18 nm
- Spacing in the middle of cells: 2 nm
- Area of the cells: $27 \times (18 \times 18) = 8748$ nm^2
- Area covered (Total): 32,964 nm^2

Figure 11 presents the simulation results for the 1:4 DeMux, showing how three different waveforms are applied to inputs $S0$, $S1$, and D. When the select lines $S1$ and $S0$ are set to 00, the input D is directed to output $Y0$. If $S1$ and $S0$ are 01, input D is routed to output $Y1$. For select Lines 10 and 11, input D appears at outputs $Y2$ and $Y3$, respectively. As shown in the results, the propagation delay from the input QCA cells to output QCA cell has been found uniform, with a delay of two clock cycles or eight phases. All outputs are accurately produced after a delay of eight clock phases, as indicated by the red boxes in Figure 11. Notably, the correct outputs appear during the hold phase of Clock 3. For clarity, the simulation results for the proposed 1:4 DeMux only shows Clock 3.

Table 4 showcases a comparative analysis between the proposed and the existing designs based on QCA cells, total layout area, circuit cost and delay. Various methodologies are available to evaluate the cost function of QCA circuits. In this work, the performance of the proposed architectures

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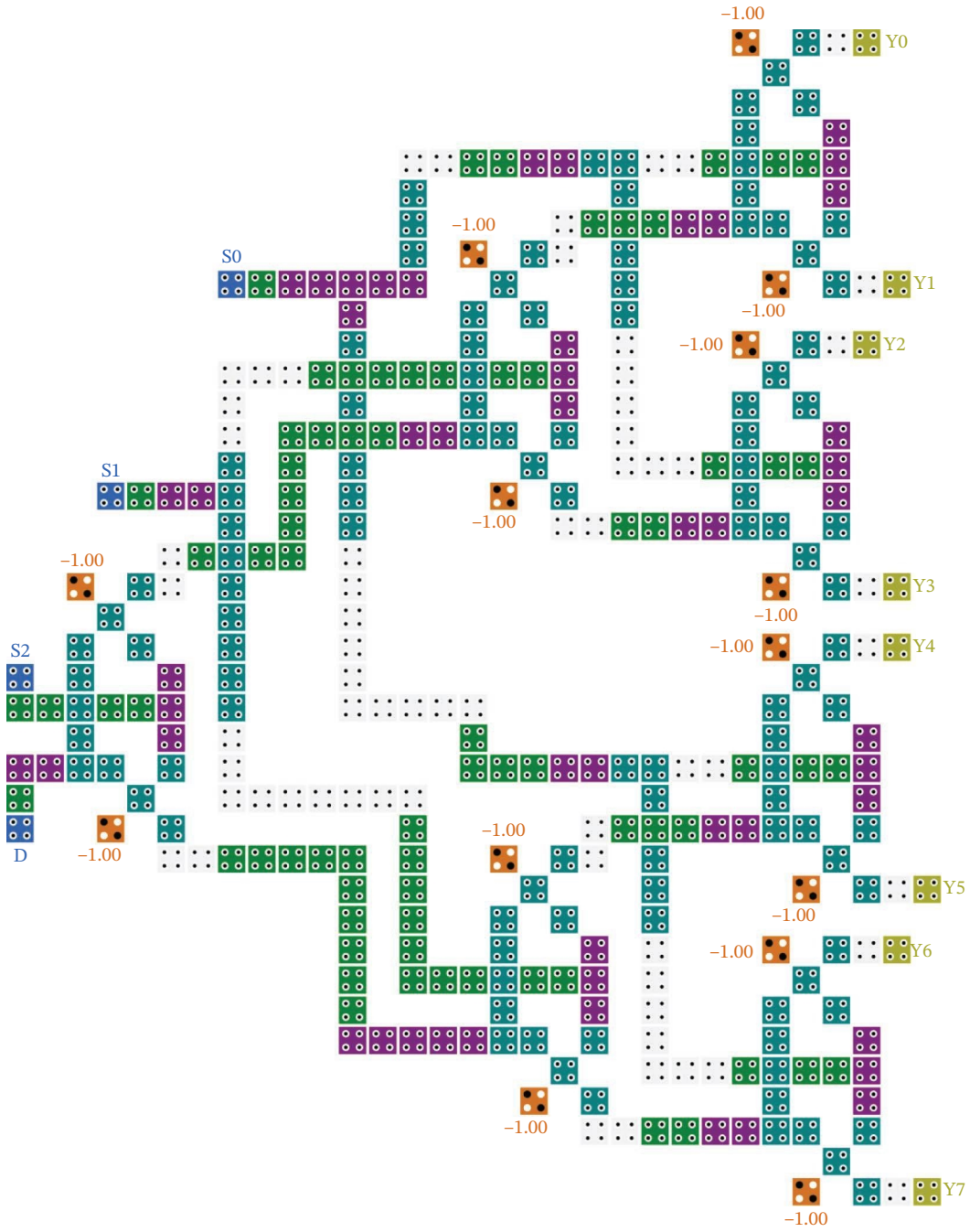


FIGURE 13 | Layout of proposed 1:8 DeMux using 1:2 DeMuxs.

is analyzed using a widely accepted evaluation metric that considers both propagation delay, represented in terms of clock cycles and physical area, measured in square micrometers (μm^2). Accordingly, the cost function is defined as $\text{cost} = (\text{area}) \times (\text{latency})^2$. Table 4 also presents parameters like input/output cell accessibility, interfacing with cascading stages and limitations. The proposed 1:2 DeMux exhibits higher standalone delay and area, it enables predictable linear scaling and lower effective overhead in multistage QCA architectures. The increased clock latency and area in the proposed design are intentional design choices made to address key limitations observed in many compact QCA DeMux architectures. Some designs reduce latency (0.25–0.5 clock cycles) and area by aggressively eliminating cells and clock zones. This frequently leads to (i) single-gate per clock zone along critical signal paths, (ii) rotated gates and vertical shifts, or (iii) inaccessible inputs/outputs. These factors negatively impact signal stability, polarization reliability and scalability of the circuit, especially when cascaded or integrated into large systems. The proposed design also involves the addition of extra clock phases needed for proper signal stabilization across various clock zones, maintaining consistent propagation delay for all outputs and avoiding congestion at a single clock zone level that may undermine reliability when operated at higher frequencies. The small increase in area follows from maintaining direct accessibility of all inputs and outputs, avoiding rotated cells and vertical shifts that require tighter fabrication tolerances, and preserving a flat, coplanar configuration

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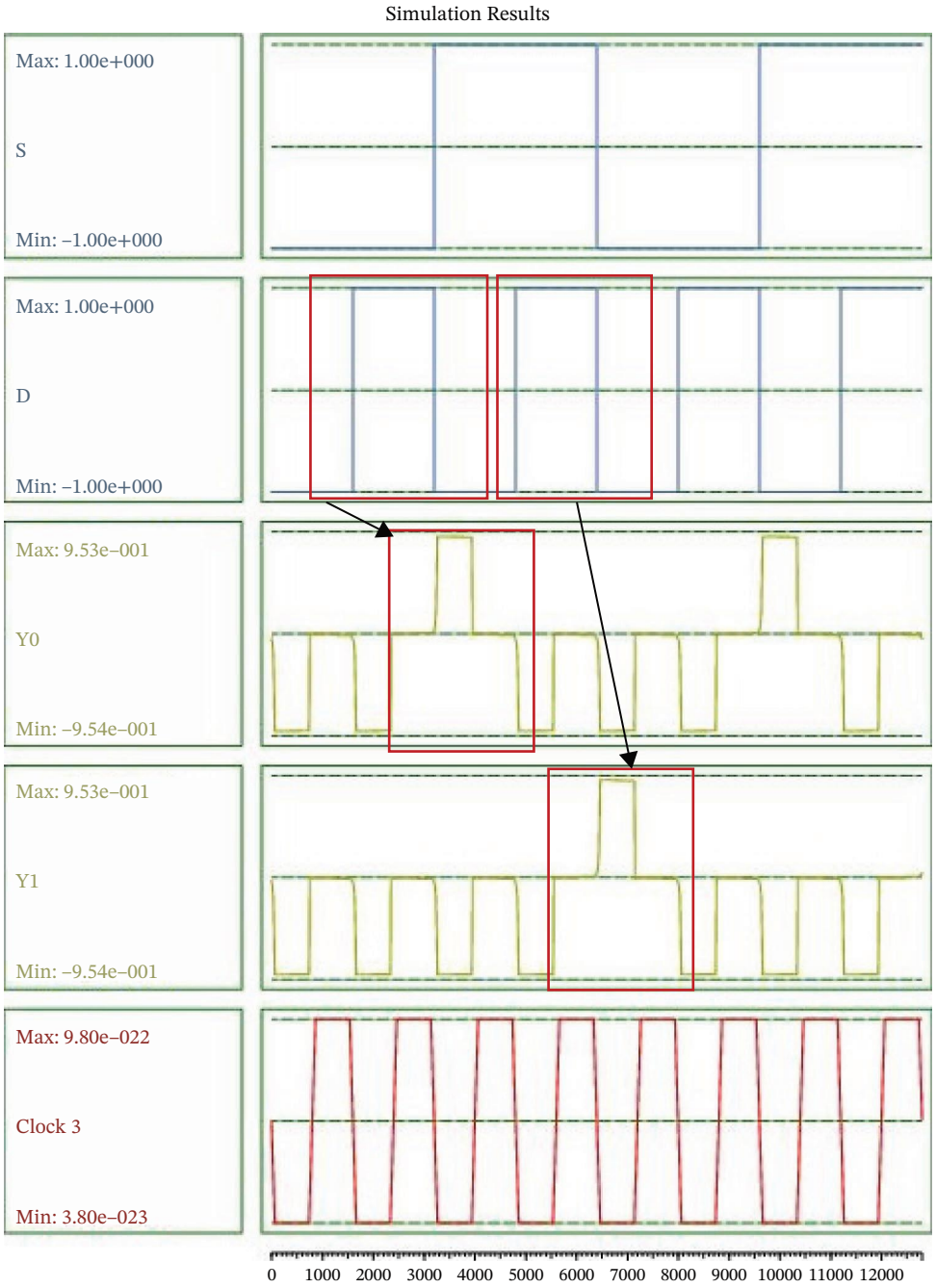


FIGURE 14 | Simulation waveform of the proposed 1:2 DeMux.

amenable to modular extension. None of the existing articles have examined the direct accessibility of input and output cells as a comparing parameter for facilitating quick interfacing with other stages. This article, however, addresses this aspect. The designs reported in [15] implemented a 1:2 DeMux using QCA technology. As outlined in Section 3, these designs significantly reduced cell count and delay by focusing on minimizing overall complexity. However, they encountered issues with direct interfacing and input/output cell accessibility. In contrast, the design in [14] addresses these issues but face other challenges, such as reliability and complexity in physical realization. Notably, most of the existing designs listed in Table 4 require substantial modifications to address these limitations, often necessitating additional 90^0 cells and multilayer/coplanar crossovers. These modifications can lead to increases in the number of cells, QCA area, cost, and overall delay. So, it is worth noting that proposed 1:2 DeMux is superior in terms of modularity, ease of interfacing, and physical realizability, rather than absolute minimization of individual performance metrics. Therefore, the proposed 1:2 DeMux structure is a suitable candidate that paves the way for designing higher-order DeMuxs with greater ease.

Implementing a 1:4 DeMux using three 1:2 DeMux modules positively impacts overall circuit performance, particularly in terms of cell count and effective area, when compared with previously reported designs. The reported metrics in Table 5 represent standalone layouts, whereas system-level usage often alters the effective

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TABLE 4 | Performance metrics comparison of existing and proposed 1:2 DeMux designs.

Design	No. of cells	Total area (μm^2)	Latency (clock cycle)	Cost= (area) × (latency)^2	All input and output cells accessible directly?	Direct interfacing possible in cascading system?	Limitation
[14]	32	0.026	1	0.026	Yes	Yes	Reliability issue as one cell placed in one clock zone
[15]	11	0.00648	0.25	0.000405	No	No	Potential interference issues as close placement of input and output cells increase intercell interaction
[16]	21	0.017	0.50	0.00425	Yes	No	Need additional cells, delay, and crossover to connect the external inputs
[17]	21	0.023	0.50	0.00575	Yes	No	Output cells are not aligned (not in same direction), need additional cells, delay, and crossover to connect the external inputs
[18]	21	0.020	0.50	0.005	No	No	Need additional cells, delay, and crossover to connect the external inputs
[19]	19	0.02	0.50	0.005	No	No	Need additional cells, delay, and crossover to connect the external inputs
[20]	20	0.023	0.50	0.00575	Yes	No	Input and output cells are not aligned (not in same direction), need additional cells, delay, and crossover to connect the external inputs
[21]	19	0.014	0.25	0.000875	Yes	No	Need additional cells, delay, and crossover to connect the external inputs

(Continues)

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TABLE 4 | (CONTINUED)

Design	No. of cells	Total area (μm^2)	Latency (clock cycle)	Cost= (area) × (latency)^2	All input and output cells accessible directly?	Direct interfacing possible in cascading system?	Limitation
[22]	9	0.0030	0.25	0.0001875	No	No	Potential interference issues as close placement of input and output cells increases inter-cell interaction, need additional cells, delay, and crossover to connect the external inputs
[23]	17	0.015	0.25	0.0009375	Yes	No	Input and output cells are not aligned (not in same direction), need additional cells and crossover to connect the external inputs.
[24]	16	0.02	0.25	0.00125	Yes	No	Input and output cells are not aligned (not in same direction), need additional cells and crossover to connect the external inputs.
[25]	10	0.0076	0.25	0.000475	No	No	Potential interference issues as close placement of input, output cells increases intercell interaction, input and output cells are not aligned, need additional cells and crossover to connect the external inputs.
Proposed	27	0.033	1	0.033	Yes	Yes	Increasing clock phases

cost of designs lacking direct I/O accessibility. Table 5 shows that the proposed structure occupies less area and a smaller number of QCA cells compared with other designs [16, 17].

In addition, the designs from [19, 21] are showing superiority in terms of cell count, area, latency, and cost compared to the proposed design. Quick observation shows, the 1:4 DeMux structure in [19] requires cell rotations and vertical shifts of some cells at crossovers, specifically by 9 nm, and no direct access to input cell I and $S0$. As a consequence, additional routing cells and coplanar or multilayer crossovers become unavoidable when extending the design to larger DeMuxs or embedding it within a system-level datapath. These necessary postlayout modifications effectively increase the actual cell count, area, and delay

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TABLE 5 | Performance metrics comparison of existing and proposed 1:4 DeMux designs.

Design	No of cells	Total area (μm^2)	Latency (clock cycle)	Cost= (area) × (latency)^2	Crossover needed	Limitation	Challenges in physical realization
[17]	187	0.18	1.75	0.55125	Coplanar	Cannot access the inputs directly. Ex. Cascaded stages. Need additional cells and crossover to connect the inputs from previous stage. Layout needs modification	Less
[16]	140	0.102	1	0.102	Multilayer	Cannot access the inputs directly. Ex. Cascaded stages. Need additional cells and crossover to connect the inputs from previous stage. Layout needs modification. Large number of cells in one clock zones.	High
[19]	92	0.12	1	0.12	Coplanar	Cannot access the inputs directly. Ex. Cascaded stages. Need additional cells and crossover to connect the inputs from previous stage. Layout needs modification. Needs rotated cells and vertical shifting of cells by 9 nm.	Medium
[21]	72	0.06	0.75	0.03375	Multilayer	Cannot access the inputs directly. Ex. Cascaded stages. Need additional cells and crossover to connect the inputs from previous stage. Layout needs modification.	High

(Continues)

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TABLE 5 | (CONTINUED)

Design	No of cells	Total area (μm^2)	Latency (clock cycle)	Cost= (area) × (latency)^2	Crossover needed	Limitation	Challenges in physical realization
[25]	64	0.0876	0.75	0.04927	Multilayer	Cannot access the inputs directly. Ex. Cascaded stages. Need additional cells, delay and crossover to connect the inputs from previous stage. Layout needs modification.	High
Proposed	105	0.16	2	0.64	Coplanar	Nil	Less

TABLE 6 | Performance metrics comparison of existing and proposed 1:8 DeMux designs.

Design	No of cells	Total area (μm^2)	Latency (clock cycle)	Cost= (area) × (latency)^2	Crossover needed	Limitation	Challenges in physical realization
[21]	206	0.06	1.50	0.135	Multilayer	Cannot access the inputs directly. Ex. Cascaded stages. Need additional cells and crossover to connect the inputs from previous stage. Layout needs modification.	High
Proposed	325	0.55	3	4.95	Coplanar	Increasing clock phases	Less

beyond the values reported for the isolated 1:4 block. Designs in [21, 25] need multilayer layout and require additional cells and crossover to connect the cascade stages. Notably, most of the existing designs listed in Table 5 require substantial modifications to address their own limitations, often necessitating additional 90^0 cells and multilayer/coplanar crossovers. These modifications can lead to increases in the number of cells, QCA area, and overall delay. In contrast, our proposed 1:4 DeMux architecture offers easier interfacing, quicker access to input and output cells, and fewer challenges in physical realization compared to existing structures.

As discussed in Section 3, there are only a few 1:8 QCA DeMuxs available in the literature. Consequently, we have selected a few notable QCA DeMuxs from the literature that could compete with the proposed design. The detailed comparison in Table 6 indicates that the multilayer design by Sharma [21] is the strongest competitor to the proposed design. However, the design is less advantageous for multistage QCA structures, as it requires additional cells and multilayer crossovers for direct interfacing. This modification can lead to increases in the number of cells, QCA area, overall delay and overall multilayer design complexity. From the previous analysis, it is concluded that the proposed DeMux is highly efficient and well-suited for designing higher-order circuits.

Higher-order DeMuxs are not implemented in [14], and therefore, these sources do not include expansions with modules. In contrast, higher-order DeMuxs have been successfully implemented using the 1:2 DeMux proposed in this work. The implementations of higher-order DeMuxs are detailed in Table 7.

QCADesigner-E tool has been used for the evaluation of energy dissipation of the proposed DeMux. The energy dissipation results are tabulated in Table 8 considering existing work [21]. The energy analysis clearly indicates that the proposed designs achieve more significant improvements in higher-order DeMux

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TABLE 7 | DeMux whether or not are enlarged with module.

DeMux	Different size DeMux			Expand?
	1:2	1:4	1:8	
[22]	✓	×	×	No
[23]	✓	×	×	No
[24]	✓	×	×	No
[25]	✓	✓	×	No
[15]	✓	×	×	No
[16]	✓	✓	×	No
[14]	✓	×	×	No
[17]	✓	✓	×	No
[18]	✓	×	×	No
[19]	✓	✓	×	No
[20]	✓	×	×	No
[21]	✓	✓	✓	Yes
Proposed	✓	✓	✓	**Yes**

Note: ✓represents DeMux is expanded with module. × represents DeMux is not expanded with module.

TABLE 8 | Energy dissipations in the presented DeMux structures.

Designs	Total energy dissipation (eV)	Error(eV)	Average energy dissipation(eV)	Error (eV)
1:2 DeMux [21]	1.23e−002	−1.26e−003	1.12e−003	−1.14e−004
1:2 DeMux proposed	7.92e−003	−7.37e−004	7.20e−004	−6.70e−005
1:4 DeMux [21]	2.83e−002	−2.59e−003	2.57e−003	−2.36e−00
1:4 DeMux proposed	1.72e−002	−1.22e−003	1.57e−003	−1.11e−004
1:8 DeMux [21]	6.02e−002	−4.96e−003	5.47e−003	−4.51e−004
1:8 DeMux proposed	5.59e−002	−3.82e−003	5.08e−003	−3.47e−004

architectures. A direct quantitative comparison of energy dissipation with all previously reported QCA-based DeMux designs is not always feasible, as most existing works do not report energy results using QCADesigner-E or do not specify identical clocking schemes, temperature assumptions, and simulation parameters. Because energy dissipation in QCA circuits is highly dependent on factors like the number of cells involved, the layout of clock zones and switching behavior commuting time (adiabatic switching), any cross-evaluation based on heterogeneous simulation environments might result in erroneous conclusions. Therefore, the energy values reported in Table 8 are intended to serve as baseline reference data for modular, coplanar 1:2^n QCA DeMux designs evaluated under consistent conditions. Qualitatively, the observed increase in energy dissipation from 1:2 to 1:8 DeMux follows the expected scaling trend with circuit size and number of clock cycles, which is consistent with energy behavior reported in related QCA logic designs.

The error value reported by QCAdesigner-E represents the numerical uncertainty associated with the energy dissipation calculation. Because the energy is computed through time-step–based numerical integration of cell polarization dynamics, a finite error is unavoidable and mainly depends on simulation parameters such as the time step size, convergence criteria, and clocking resolution. In our simulations, the reported error is significantly smaller than the corresponding total energy dissipation values, indicating good numerical convergence and reliability of the results. Therefore, the error does not affect the qualitative or quantitative comparison between different QCA designs presented in this work. The error in QCAdesigner-E energy analysis quantifies the numerical reliability of energy dissipation results low error means trustworthy energy values; high error means the results should be treated with caution.

6 | Conclusion

The current article introduces a new, simplified layout for a 1:2 DeMux utilizing QCA-based RMGs. The key advantage of this new design is its straight forward layout and modular structure, which enables easy integration into various networks. This structure serves as a foundational unit for constructing higher-order

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DeMuxs. Consequently, coplanar 1:4 and 1:8 DeMuxs were realized by this method. The layouts are verified via QCA Designer simulator with performance analysis. The DeMux for 1:2, 1:4 and 1:8 comprises of 27, 105, and 325 cells, respectively, with respective delays of one clock cycle, two clock cycles, three clock cycles. These results support a path towards architecture modularization, adaptation, and realization that achieves significant improvements in terms relative to local minimization of performance criteria. These benefits are seen in the design assumptions and simulation framework that have been adopted. Although the results demonstrate that the proposed architecture includes all basic logic components for nanoscale DeMux design, other practical aspects of implementation like fabrication limitations, fault tolerance, and large-scale integration are considerations for future research. This innovation has the potential to enable a new generation of DeMux-based circuits in suitable design environments such as nanocommunication networks.

Author Contributions

Paramjeet: conceptualization, methodology, design, and simulation. Saptarshi Gupta: supervision and validation. Rupali Singh: review and editing. Trailokya Nath Sasamal: technical review and revision. Aditya Agarwal: data curation. Minakshi Sanadhya: literature review and documentation. Dharmbir Prasad: review and editing. Srinivasan S. Balapangu: framework improvement and manuscript revision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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